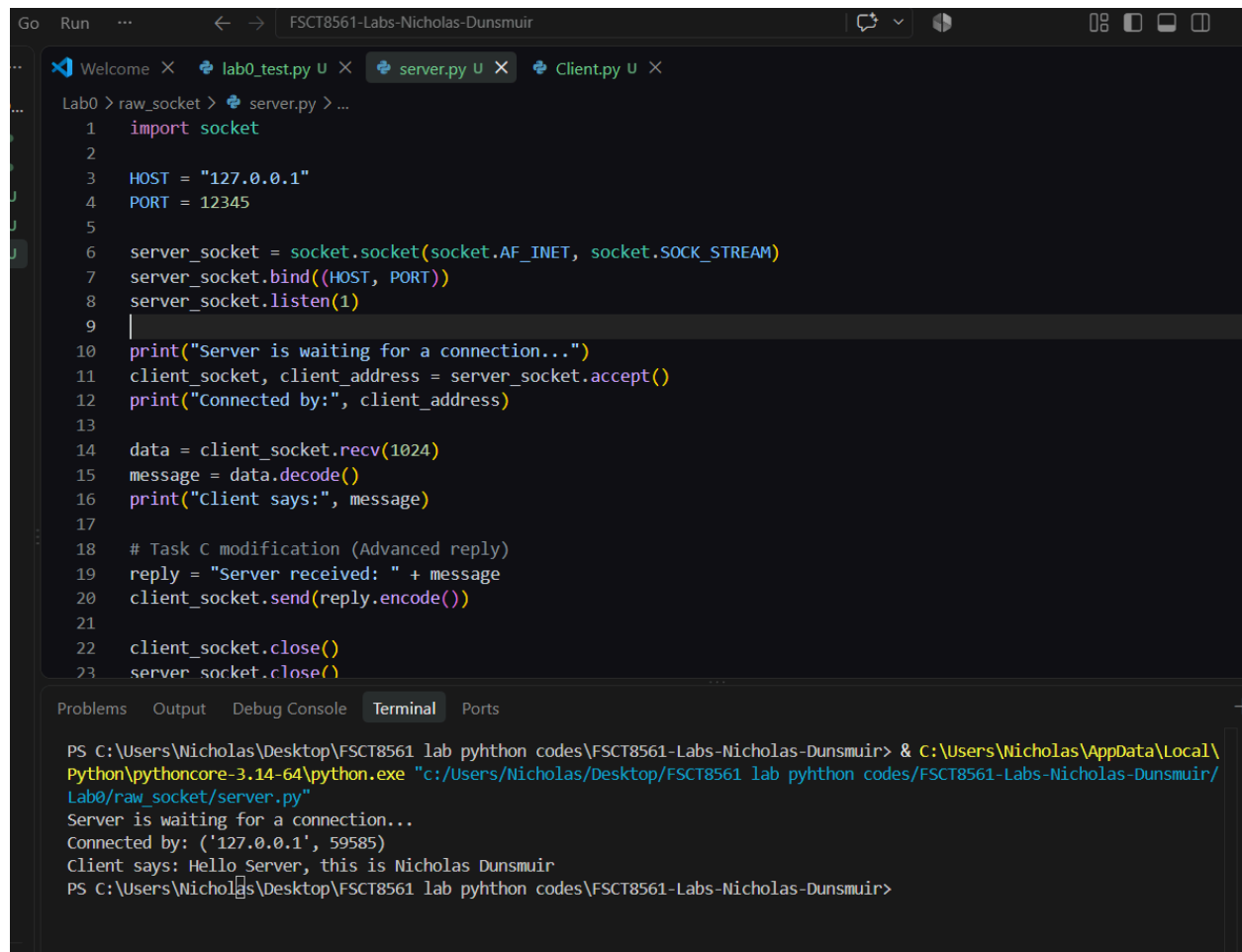


Lab0

Screenshots of successful server and client interaction in the raw_socket folder:

SERVER



The screenshot shows a VS Code editor window with the file explorer on the left displaying the 'raw_socket' folder. The editor has four tabs: 'Welcome', 'lab0_test.py', 'server.py', and 'Client.py'. The 'server.py' tab is active, showing a Python script that listens on 127.0.0.1:12345, accepts a connection, receives a message, and sends a reply. The terminal at the bottom shows the command to run the script and its output, which matches the script's print statements.

```
Go Run ... FSCT8561-Labs-Nicholas-Dunsmuir
Welcome X lab0_test.py U X server.py U X Client.py U X
Lab0 > raw_socket > server.py > ...
1 import socket
2
3 HOST = "127.0.0.1"
4 PORT = 12345
5
6 server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
7 server_socket.bind((HOST, PORT))
8 server_socket.listen(1)
9
10 print("Server is waiting for a connection...")
11 client_socket, client_address = server_socket.accept()
12 print("Connected by:", client_address)
13
14 data = client_socket.recv(1024)
15 message = data.decode()
16 print("Client says:", message)
17
18 # Task C modification (Advanced reply)
19 reply = "Server received: " + message
20 client_socket.send(reply.encode())
21
22 client_socket.close()
23 server_socket.close()
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir> & C:\Users\Nicholas\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSC8561 lab pyhton codes/FSC8561-Labs-Nicholas-Dunsmuir/Lab0/raw_socket/server.py"
Server is waiting for a connection...
Connected by: ('127.0.0.1', 59585)
Client says: Hello Server, this is Nicholas Dunsmuir
PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir>
```

CLIENT

```
Lab0 / raw_socket / Client.py / ...
1  import socket
2
3  HOST = "127.0.0.1"
4  PORT = 12345
5
6  client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
7  client_socket.connect((HOST, PORT))
8
9  message = "Hello Server, this is Nicholas Dunsmuir" # Task A modification
10 client_socket.send(message.encode())
11
12 response = client_socket.recv(1024)
13 print("Server replied:", response.decode())
14
15 client_socket.close()
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir> & C:\Users\Nicholas\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSC8561 lab pyhton codes/FSC8561-Labs-Nicholas-Dunsmuir/Lab0/raw_socket/server.py"
Server is waiting for a connection...
Connected by: ('127.0.0.1', 59585)
Client says: Hello Server, this is Nicholas Dunsmuir
PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir>
```

HTML Code output:

```
PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir> & C:\Users\Nicholas\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSC8561 lab pyhton codes/FSC8561-Labs-Nicholas-Dunsmuir/Lab0/raw_socket/http_client.py"
HTTP/1.1 200 OK
Date: Thu, 10 Sep 2026 17:40:45 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Last-Modified: Sun, 30 Aug 2026 04:11:49 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 1702
cf-cache-status: HIT
CF-RAY: a390383a3fff58e5-YVR

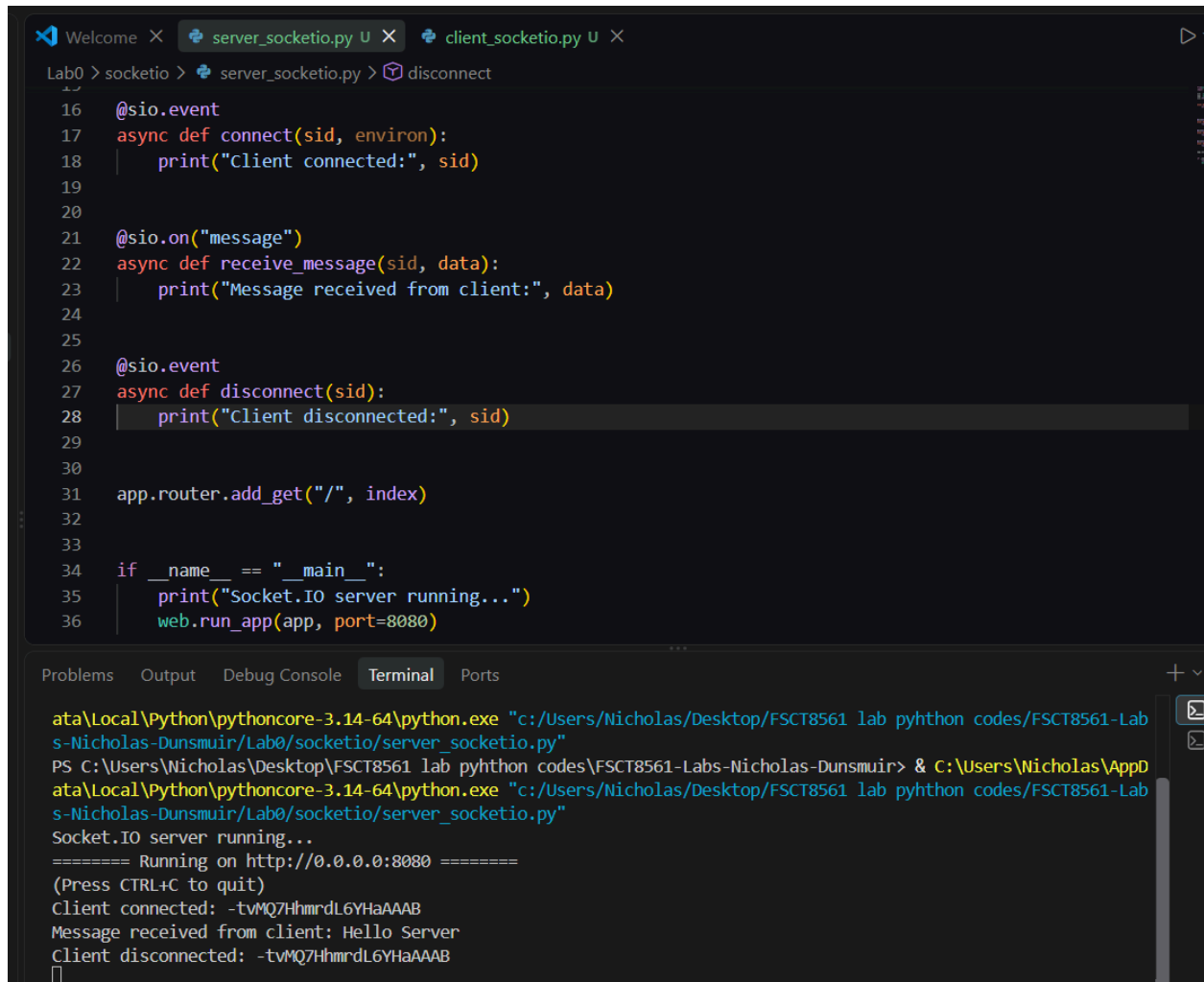
22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

0

PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir>
```

Socket challenge lab section screenshots:

SERVER:



The image shows a Visual Studio Code editor window with two tabs: 'server_socketio.py' and 'client_socketio.py'. The 'server_socketio.py' tab is active, displaying a Python script that uses the Socket.IO library. The script defines three asynchronous event handlers: 'connect' (prints 'Client connected:'), 'message' (prints 'Message received from client:'), and 'disconnect' (prints 'Client disconnected:'). It also sets up a Flask-like web application with a root route and runs it on port 8080.

```
16 @sio.event
17 async def connect(sid, environ):
18     print("Client connected:", sid)
19
20
21 @sio.on("message")
22 async def receive_message(sid, data):
23     print("Message received from client:", data)
24
25
26 @sio.event
27 async def disconnect(sid):
28     print("Client disconnected:", sid)
29
30
31 app.router.add_get("/", index)
32
33
34 if __name__ == "__main__":
35     print("Socket.IO server running...")
36     web.run_app(app, port=8080)
```

Below the editor, the 'Terminal' panel shows the command prompt output. It displays the command to run the server script, followed by the server's startup message and the log of a client connection and message exchange.

```
ata\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSCT8561 lab pyhton codes/FSCT8561-Lab
s-Nicholas-Dunsmuir/Lab0/socketio/server_socketio.py"
PS C:\Users\Nicholas\Desktop\FSCT8561 lab pyhton codes\FSCT8561-Labs-Nicholas-Dunsmuir> & C:\Users\Nicholas\AppData
ata\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSCT8561 lab pyhton codes/FSCT8561-Lab
s-Nicholas-Dunsmuir/Lab0/socketio/server_socketio.py"
Socket.IO server running...
===== Running on http://0.0.0.0:8080 =====
(Press CTRL+C to quit)
Client connected: -tvMQ7HhmrDL6YHaAAAB
Message received from client: Hello Server
Client disconnected: -tvMQ7HhmrDL6YHaAAAB
```

CLIENT:

The image shows a Visual Studio Code editor with two tabs: 'server_socketio.py' and 'client_socketio.py'. The 'client_socketio.py' tab is active, displaying a Python script. The script defines two event handlers: 'connect()' which prints 'Connection established', and 'disconnect()' which prints 'Disconnected from server'. It then connects to 'http://localhost:8080', sends a message 'Hello Server', sleeps for 1 second, and disconnects. Below the code, the 'Terminal' panel shows the execution of the script. It displays a warning about the 'websocket-client' package, followed by the successful connection, message sending, and disconnection.

```
Lab0 > socketio > client_socketio.py > ...
6
7 @sio.event
8 def connect():
9     print("Connection established")
10
11
12 @sio.event
13 def disconnect():
14     print("Disconnected from server")
15
16
17 sio.connect("http://localhost:8080")
18
19 message = "Hello Server"
20 print("Sending message:", message)
21
22 sio.emit("message", message)
23
24 time.sleep(1)
25
26 sio.disconnect()
```

Problems Output Debug Console Terminal Ports

socketio.exceptions.ConnectionError: HTTPConnectionPool(host='localhost', port=8080): Max retries exceeded with url: /socket.io/?transport=polling&EIO=4&t=1789062869.9735916 (Caused by NewConnectionError("HTTPConnection(host='localhost', port=8080): Failed to establish a new connection: [WinError 10061] No connection could be made because the target machine actively refused it"))

PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir> & C:\Users\Nicholas\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Nicholas/Desktop/FSC8561 lab pyhton codes/FSC8561-Labs-Nicholas-Dunsmuir/Lab0/socketio/client_socketio.py"

websocket-client package not installed, only polling transport is available

Connection established

Sending message: Hello Server

Disconnected from server

PS C:\Users\Nicholas\Desktop\FSC8561 lab pyhton codes\FSC8561-Labs-Nicholas-Dunsmuir>

Reflection questions:

- 1: a) A socket is an endpoint used by an application to communicate over a network
 - b) AF_INET means an internet ipv4 connection for the socket
 - c) Sock_Stream is specifying TCP
-
- 2: a) the host connected to example.com
 - b) Port 80 was used
 - c) Its the standard HTTP port
 - d) the raw HTTP response the content and headers from the server

Security Reflection Questions:

1. It risks unauthorized access, denial-of-service attacks, and malicious interactions from untrusted sources.

2. No, raw sockets send unencrypted data
3. Yes, because the data is unencrypted
4. To prevent application vulnerabilities and injection attacks
5. To verify the identity of the client connecting to the server and ensure only authorized entities have access
6. Implement encryption and add client authentication

In this lab I familiarized myself with the client-server communication using python sockets and how to make a basic TCP connection that included sending a single message, receiving a response and then disconnecting. The initial configuration for the challenge section of the lab had me stumped on why the connection wasn't working. I then figured out that I needed to install additional addons to the environment (PIP) to facilitate the connection between the server and client.

LINK TO PUBLIC GITHUB REPOSITORY:

<https://github.com/NDunsmuir-BCIT/FSCT8561-Labs-Nicholas-Dunsmuir>